

Phase 1: Project Overview, Requirement Analysis & Planning

Project: Auto Ticket Classification using Flow Designer | **Platform:** ServiceNow | **Domain:** IT Service Management (ITSM) Automation

1. Executive Summary & Problem Statement

The educational institution's School IT Helpdesk receives scores of support requests every day from faculty members, administrative staff, and students. These incoming incidents span across diverse operational domains—campus Wi-Fi connectivity blackouts, smart projector hardware breakdowns in lecture halls, forgotten password and account lockouts on student portals, and workstation slowdowns.

Historically, the IT desk team manually examined each submitted ticket, diagnosed the issue type, assigned proper Category and Subcategory attributes, and routed tickets to technicians. This manual intervention resulted in critical operational bottlenecks:

- **Substantial Latency:** Critical classrooms sat idle waiting 30 to 90 minutes solely for manual categorization and triage.
- **Human Inconsistency:** Misclassified tickets were frequently routed to incorrect support groups, multiplying resolution hops and extending Mean Time to Resolve (MTTR).
- **Zero Immediate Feedback:** Callers received no immediate automated confirmation that their request was logged, categorized, and assigned.

2. Project Objectives & Success Criteria

- **Automated Ingestion & Classification:** Instantly inspect short description and description keywords upon ticket creation and dynamically populate Category and Subcategory fields.
- **Eliminate Manual Triaging:** Remove human classification overhead for over 80% of standard campus IT incidents.
- **Zero-Script Scalability:** Implement a maintainable, robust, audit-friendly, and upgrade-safe solution completely within ServiceNow Flow Designer (No-Code approach).
- **Instant Caller Acknowledgement:** Trigger automated transactional notification emails to callers immediately upon ticket generation and classification.

3. Detailed Functional & Non-Functional Requirements

The functional and technical requirements for Phase 1 are detailed in the matrix below:

Requirement Category	Description	ServiceNow Component
Automatic Categorization	Identify keywords (e.g., 'WiFi', 'Projector', 'Password', 'Slow') and auto-set category & subcategory without manual click.	Flow Designer Trigger & Logic Blocks

Requirement Category	Description	ServiceNow Component
Hierarchical Integrity	Strict field dependency enforcement: Subcategory values must dynamically filter strictly based on selected Category.	Data Dictionary Dependent Field Mapping
Communication Loop	Instant notification email sent to caller's registered email with ticket number and recorded details.	Flow Designer Action: Send Email
Data Traceability & Governance	All development artifacts packaged into an isolated Update Set for seamless audit and deployment governance.	System Update Sets (Local Update Set)

4. High-Level Architecture & Lifecycle Flow

The automated ticket lifecycle flows through five distinct stages:

1. Student or Faculty Member submits an IT request via the portal or form with Caller Name, Short Description, and Description.
2. ServiceNow Database Trigger fires on record insert (Table: u_incident_workflow, Condition: Category IS EMPTY).
3. Flow Designer Automation Engine evaluates keyword decision blocks (Wi-Fi, Projector, Password, Slow Computer).
4. Ticket fields (Category, Subcategory, State) are automatically updated on the instance database.
5. Automated confirmation email is dynamically dispatched to the caller's registered institutional address.

5. Implementation Activities & Configuration Plan

Activity 1.1: Business Environment & Use Case Mapping

Establish exact operational parameters for the School IT Helpdesk. Incoming tickets are systematically mapped into four core operational categories: Network (Wi-Fi), Hardware (Projector), Access (Forgot Password), and Performance (Slow Computer).

Activity 1.2: Development Governance & Update Set Creation

All customizations must be strictly tracked within a dedicated ServiceNow Local Update Set to ensure transportability between Dev, Test, and Production instances.

1. Navigate to: All Menus → System Update Sets → Local Update Sets.
2. Click the 'New' button.
3. Set Name: 'Project Update Set' (or 'Auto Ticket Classification Update Set').
4. Set Application: Global.
5. Set State: In progress.
6. Click 'Submit and Make Current'.

Outcome: Every subsequent dictionary modification, form design alteration, and flow creation is automatically recorded into this update set container for version control and migration.